

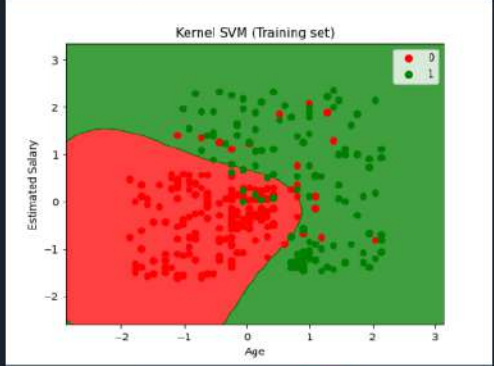
Spyder (Python 3.12)

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C:\Users\Lenovo\grid_search.py

```
1 # Grid Search
2
3 # Importing the libraries
4 import numpy as np
5 import matplotlib.pyplot as plt
6 import pandas as pd
7
8 # Importing the dataset
9 dataset = pd.read_csv(r"C:\Users\Lenovo\OneDrive\Desktop\Naresh i Note\sept\12th-09\10th - Cross validation\10th - Cross vali
10 X = dataset.iloc[:, 2:4].values
11 y = dataset.iloc[:, -1].values
12
13 # Feature Scaling
14 from sklearn.preprocessing import StandardScaler
15 sc = StandardScaler()
16 X = sc.fit_transform(X)
17
18 # Splitting the dataset into the Training set and Test set
19 from sklearn.model_selection import train_test_split
20 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.25, random_state = 0)
21
22 # Training the Kernel SVM model on the Training set
23 from sklearn.svm import SVC
24 classifier = SVC()
25 classifier.fit(X_train, y_train)
26
27 # Predicting the Test set results
28 y_pred = classifier.predict(X_test)
29
30 # Making the Confusion Matrix
31 from sklearn.metrics import confusion_matrix
32 cm = confusion_matrix(y_test, y_pred)
33 print(cm)
34
35 from sklearn.metrics import accuracy_score
36 ac = accuracy_score(y_test, y_pred)
37 print(ac)
38
39 bias = classifier.score(X_train, y_train)
```

Kernel SVM (Training set)



Help Variable Explorer Plots Files Code Analysis

Console 1/A X

```
In [2]: bias
Out[2]: 0.9066666666666665

In [3]: variance
Traceback (most recent call last):

  Cell In[3], line 1
    variance
NameError: name 'variance' is not defined

In [4]: variance
Out[4]: 0.93

In [5]: runfile('C:/Users/Lenovo/untitled1.py', wdir='C:/Users/
Lenovo')
[[64  4]
 [ 3 29]]
0.93
```

Python Console History

conda (Python 3.12.7) Completions: conda LSP: Python Line 9, Col 210 UTF-8 CRLF RW Mem 90%

Spyder (Python 3.12)

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C:\Users\Lenovo\grid_search.py

py

svr.py

flask01.py

L1,L2_Reg.py

lgistic_reg.py

naive_bayes.py

decision_tree_classification.py

k_fold_cross_validation.py

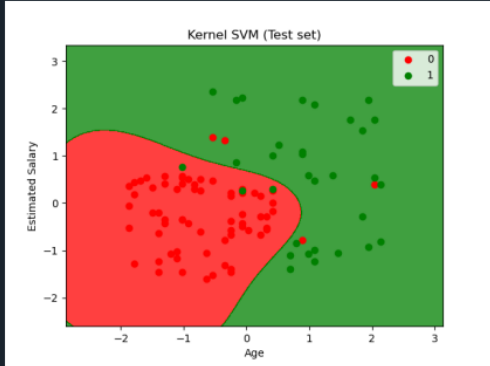
grid_search.py

```
58 grid_search = grid_search.fit(X_train, y_train)
59 best_accuracy = grid_search.best_score_
60 best_parameters = grid_search.best_params_
61 print("Best Accuracy: {:.2f} %".format(best_accuracy*100))
62 print("Best Parameters:", best_parameters)
63
64 # randomized search cv
65
66
67 # Visualising the Training set results
68 from matplotlib.colors import ListedColormap
69 X_set, y_set = X_train, y_train
70 X1, X2 = np.meshgrid(np.arange(start = X_set[:, 0].min() - 1, stop = X_set[:, 0].max() + 1, step = 0.01),
71                     np.arange(start = X_set[:, 1].min() - 1, stop = X_set[:, 1].max() + 1, step = 0.01))
72 plt.contourf(X1, X2, classifier.predict(np.array([X1.ravel(), X2.ravel()]).T).reshape(X1.shape),
73             alpha = 0.75, cmap = ListedColormap(('red', 'green')))
74 plt.xlim(X1.min(), X1.max())
75 plt.ylim(X2.min(), X2.max())
76 for i, j in enumerate(np.unique(y_set)):
77     plt.scatter(X_set[y_set == j, 0], X_set[y_set == j, 1],
78               c = ListedColormap(('red', 'green'))(i), label = j)
79 plt.title('Kernel SVM (Training set)')
80 plt.xlabel('Age')
81 plt.ylabel('Estimated Salary')
82 plt.legend()
83 plt.show()
84
85 # Visualising the Test set results
86 from matplotlib.colors import ListedColormap
87 X_set, y_set = X_test, y_test
88 X1, X2 = np.meshgrid(np.arange(start = X_set[:, 0].min() - 1, stop = X_set[:, 0].max() + 1, step = 0.01),
89                     np.arange(start = X_set[:, 1].min() - 1, stop = X_set[:, 1].max() + 1, step = 0.01))
90 plt.contourf(X1, X2, classifier.predict(np.array([X1.ravel(), X2.ravel()]).T).reshape(X1.shape),
91             alpha = 0.75, cmap = ListedColormap(('red', 'green')))
92 plt.xlim(X1.min(), X1.max())
93 plt.ylim(X2.min(), X2.max())
94 for i, j in enumerate(np.unique(y_set)):
95     plt.scatter(X_set[y_set == j, 0], X_set[y_set == j, 1],
96               c = ListedColormap(('red', 'green'))(i), label = j)
97 plt.title('Kernel SVM (Test set)')
```

C:\Users\Lenovo

65 %

Kernel SVM (Test set)



Help Variable Explorer Plots Files Code Analysis

Console 1/A

c:\users\lenovo\untitled1.py:77: UserWarning: *c* argument looks like a single numeric RGB or RGBA sequence, which should be avoided as value-mapping will have precedence in case its length matches with *x* & *y*. Please use the *color* keyword-argument or provide a 2D array with a single row if you intend to specify the same RGB or RGBA value for all points.
plt.scatter(X_set[y_set == j, 0], X_set[y_set == j, 1],

Important

Figures are displayed in the Plots pane by default. To make them also appear inline in the console, you need to uncheck "Mute inline plotting" under the options menu of Plots.

c:\users\lenovo\untitled1.py:95: UserWarning: *c* argument looks like a single numeric RGB or RGBA sequence, which should be avoided as value-mapping will have precedence in case its length matches with *x* & *y*. Please use the *color* keyword-argument or provide a 2D array with a single row if you intend to specify the same RGB or RGBA value

Python Console History

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